

Classifying when p -approximate Bessel sequences expand to p -ASFs

FA/Banach run packet for arXiv:2102.03288

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Source question

The source paper is K. Mahesh Krishna and P. Sam Johnson, *Expansion of approximate Bessel sequences to approximate Schauder frames for Banach spaces*, arXiv:2102.03288. After defining p -approximate Bessel sequences, the authors ask Question 2.12:

For which classes of Banach spaces, p -approximate Bessel sequences can be expanded to a p -ASF?

space and the standard separable Hilbert space $\ell^2(\mathbb{N})$. We now state the definition of a p -approximate Bessel sequence and end the paper with an open question and a partial answer for it.

Definition 2.11. Let $p \in [1, \infty)$. An approximate Bessel sequence $(\{f_n\}_n, \{\tau_n\}_n)$ for $\mathcal{X}$ is said to be a p -approximate Bessel sequence for $\mathcal{X}$ if both the maps $\theta_f : \mathcal{X} \ni x \mapsto \theta_f x := \{f_n(x)\}_n \in \ell^p(\mathbb{N})$ and $\theta_\tau : \ell^p(\mathbb{N}) \ni \{a_n\}_n \mapsto \theta_\tau \{a_n\}_n := \sum_{n=1}^{\infty} a_n \tau_n \in \mathcal{X}$ are well-defined bounded linear operators.

Question 2.12. For which classes of Banach spaces, p -approximate Bessel sequences can be expanded to a p -ASF?

Theorem 2.13. Let $p \in [1, \infty)$. If $(\{f_n\}_n, \{\tau_n\}_n)$ is a p -approximate Bessel sequence for $\ell^p(\mathbb{N})$, then $(\{f_n\}_n, \{\tau_n\}_n)$ can be expanded to a p -ASF.

Here $1 \leq p < \infty$. A p -approximate Bessel sequence, abbreviated p -ABS below, is a pair $(\{f_n\}, \{\tau_n\}) \subseteq \mathcal{X}^* \times \mathcal{X}$ for which the analysis and synthesis maps

$$\theta_f : \mathcal{X} \rightarrow \ell^p(\mathbb{N}), \quad \theta_f x = (f_n(x))_n, \quad \text{and} \quad \theta_\tau : \ell^p(\mathbb{N}) \rightarrow \mathcal{X}, \quad \theta_\tau(a_n) = \sum_{n=1}^{\infty} a_n \tau_n$$

are bounded. It is a p -approximate Schauder frame, abbreviated p -ASF, when additionally the frame operator $S_{f,\tau} = \theta_\tau \theta_f$ is invertible.

Proof intuition

There are two separate issues in the source question. First, if the space already has one p -ASF, then any p -ABS can be completed: use a Parseval p -ASF (a_n, ω_n) for the identity, and add the vectors $(I - S_{f,\tau})\omega_n$ with the same functionals a_n . The added frame operator is exactly $I - S_{f,\tau}$, so the enlarged pair has frame operator I .

Second, the existence of one p -ASF is already known to be an operator-factorization condition through ℓ^p . The characterization theorem of Krishna–Johnson, arXiv:2010.10514, says a p -ASF is the same as bounded maps $U : \mathcal{X} \rightarrow \ell^p(\mathbb{N})$ and $V : \ell^p(\mathbb{N}) \rightarrow \mathcal{X}$ with VU invertible. That is equivalent to saying that $\mathcal{X}$ is isomorphic to a complemented subspace of ℓ^p .

Theorem 2.5. A pair $(\{f_n\}_n, \{\tau_n\}_n)$ is a p -ASF for $\mathcal{X}$ if and only if

$$f_n = h_n U, \quad \tau_n = V e_n, \quad \forall n \in \mathbb{N},$$

where $U : \mathcal{X} \rightarrow \ell^p(\mathbb{N})$, $V : \ell^p(\mathbb{N}) \rightarrow \mathcal{X}$ are bounded linear operators such that VU is bounded invertible.

Proof. ($\Leftarrow$) Clearly θ_f and θ_τ are bounded linear operators. Now let $x \in \mathcal{X}$. Then

$$(10) \quad S_{f,\tau}x = \sum_{n=1}^{\infty} f_n(x)\tau_n = \sum_{n=1}^{\infty} h_n(Ux)Ve_n = V \left(\sum_{n=1}^{\infty} h_n(Ux)e_n \right) = VUx.$$

Hence $S_{f,\tau}$ is bounded invertible.

($\Rightarrow$) Define $U := \theta_f$, $V := \theta_\tau$. Then $h_n Ux = h_n \theta_f x = h_n(\{f_k(x)\}_k) = f_n(x)$, $\forall x \in \mathcal{X}$, $Ve_n = \theta_\tau e_n = \tau_n$, $\forall n \in \mathbb{N}$ and $VU = \theta_\tau \theta_f = S_{f,\tau}$ which is bounded invertible. $\square$

The full answer

Theorem 1. Let $1 \leq p < \infty$ and let $\mathcal{X}$ be a Banach space. The following are equivalent.

1. Every p -approximate Bessel sequence for $\mathcal{X}$ can be expanded to a p -approximate Schauder frame for $\mathcal{X}$.
2. $\mathcal{X}$ admits a p -approximate Schauder frame.
3. $\mathcal{X}$ is isomorphic to a complemented subspace of $\ell^p(\mathbb{N})$.

Thus the answer to Question 2.12 in arXiv:2102.03288 is: exactly the Banach spaces isomorphic to complemented subspaces of $\ell^p(\mathbb{N})$.

Proof. We first prove $(1) \Leftrightarrow (2)$.

Assume (1). The zero pair $f_n = 0$, $\tau_n = 0$ is a p -ABS for $\mathcal{X}$. By (1), it can be expanded to a p -ASF, so $\mathcal{X}$ admits a p -ASF.

Conversely, assume that $\mathcal{X}$ admits a p -ASF. Replacing it by its canonical Parseval version, we may choose a p -ASF $(\{a_n\}, \{\omega_n\})$ whose frame operator is $I_{\mathcal{X}}$. Explicitly, if $(\{g_n\}, \{\omega_n\})$ has frame operator R , then $(\{g_n R^{-1}\}, \{\omega_n\})$ is a Parseval p -ASF.

Let $(\{f_n\}, \{\tau_n\})$ be an arbitrary p -ABS for $\mathcal{X}$, with bounded frame operator

$$S := S_{f,\tau} = \theta_\tau \theta_f.$$

Define

$$h_n := a_n, \quad \rho_n := (I_{\mathcal{X}} - S)\omega_n, \quad n \in \mathbb{N}.$$

The added analysis map is $\theta_a : \mathcal{X} \rightarrow \ell^p(\mathbb{N})$, which is bounded. The added synthesis map is

$$\theta_\rho = (I_{\mathcal{X}} - S)\theta_\omega : \ell^p(\mathbb{N}) \rightarrow \mathcal{X},$$

which is bounded because θ_ω and $I_{\mathcal{X}} - S$ are bounded. Therefore the enlarged pair is still a p -ABS. Its frame operator is

$$S_{(f,h),(\tau,\rho)} = S_{f,\tau} + S_{h,\rho} = S + (I_{\mathcal{X}} - S)S_{a,\omega} = S + (I_{\mathcal{X}} - S)I_{\mathcal{X}} = I_{\mathcal{X}}.$$

Hence the enlarged pair is a Parseval p -ASF. This proves (1).

It remains to prove $(2) \Leftrightarrow (3)$. This is the standard operator characterization of p -ASFs from Krishna–Johnson, arXiv:2010.10514, Theorem 2.5: a pair is a p -ASF for $\mathcal{X}$ if and only if there are bounded operators

$$U : \mathcal{X} \rightarrow \ell^p(\mathbb{N}), \quad V : \ell^p(\mathbb{N}) \rightarrow \mathcal{X}$$

such that VU is bounded invertible, with $f_n = h_n U$ and $\tau_n = V e_n$, where (e_n) is the unit vector basis of $\ell^p(\mathbb{N})$ and (h_n) is its coordinate functional sequence.

If $\mathcal{X}$ has a p -ASF, choose such U, V , and put $A = VU$. Then

$$J := UA^{-1} : \mathcal{X} \rightarrow \ell^p(\mathbb{N})$$

is an isomorphic embedding and $VJ = I_{\mathcal{X}}$. Consequently $JV : \ell^p(\mathbb{N}) \rightarrow \ell^p(\mathbb{N})$ is a bounded projection onto $J(\mathcal{X})$, so $J(\mathcal{X})$ is complemented in $\ell^p(\mathbb{N})$, and $\mathcal{X}$ is isomorphic to that complemented subspace.

Conversely, suppose $\mathcal{X}$ is isomorphic to a complemented subspace of $\ell^p(\mathbb{N})$. Then there are bounded operators $J : \mathcal{X} \rightarrow \ell^p(\mathbb{N})$ and $V : \ell^p(\mathbb{N}) \rightarrow \mathcal{X}$ such that J is an isomorphic embedding and $VJ = I_{\mathcal{X}}$. Define

$$f_n = h_n J, \quad \tau_n = V e_n, \quad n \in \mathbb{N}.$$

The analysis map is J , the synthesis map is V , and the frame operator is $VJ = I_{\mathcal{X}}$. Thus $(\{f_n\}, \{\tau_n\})$ is a Parseval p -ASF for $\mathcal{X}$. This proves (2), and completes the classification. $\square$

Verification and limitations

The expansion construction is a direct operator computation. The only external theorem used is the p -ASF characterization from arXiv:2010.10514, Theorem 2.5, which is cited and included as ‘supporting_paper_2010.10514.pdf’. The proof also shows that whenever expansion is possible, it can be done so that the enlarged p -ASF is Parseval.

This packet answers Question 2.12 as stated. It does not address the separate earlier source question about finite tight expansion of translation ASFs in $L^p(\mathbb{R}^d)$.

Novelty check

The cheap run indexes were searched for arXiv:2102.03288, the source title, p -approximate Bessel sequences, p -ASFs, expansion to p -ASFs, and complemented subspaces of ℓ^p . They showed related p -ASF packets for other papers but no duplicate of this classification. The source itself gives the case $\mathcal{X} = \ell^p$. The supporting paper arXiv:2010.10514 gives the operator characterization of p -ASFs but does not state this expansion classification for the later p -ABS question. Exact-pharse web searches for the source question and the complemented- ℓ^p answer did not surface a later explicit answer.

References

- K. Mahesh Krishna and P. Sam Johnson, *Expansion of approximate Bessel sequences to approximate Schauder frames for Banach spaces*. arXiv:2102.03288.
- K. Mahesh Krishna and P. Sam Johnson, *Towards characterizations of approximate Schauder frame and its duals for Banach spaces*. arXiv:2010.10514. Used for Theorem 2.5, the operator characterization of p -ASFs.

Human-review recommendation

Review as a likely valid full solution. The main checks are that the source's notion of expansion permits adding countably many pairs, as in its own Theorem 2.3 construction, and that the standard p-ASF definition is exactly the one used in arXiv:2010.10514.